

## Use of the DFT in Linear Filtering

$$x(n) = 0, \quad n < 0 \text{ and } n \geq L$$

طول L

$$h(n) = 0, \quad n < 0 \text{ and } n \geq M$$

طول M

$$x(n) \rightarrow \boxed{h(n)} \rightarrow y(n)$$

$$y(n) = \sum_{k=0}^{M-1} h(k)x(n-k) = x(n) * h(n)$$

مجموع دو سیگنال

خروجی : کانولوشن خطی

the duration of  $y(n)$  is  $L + M - 1$

DFT

DTFT

نمونه برداری در فرکانس

$$Y(k) \equiv Y(\omega)|_{\omega=2\pi k/N} \quad k = 0, 1, \dots, N-1$$

$$= X(\omega)H(\omega)|_{\omega=2\pi k/N} \quad k = 0, 1, \dots, N-1$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

N نقطه ای

$$Y(k) = X(k)H(k) \quad k = 0, 1, \dots, N-1$$

DFT

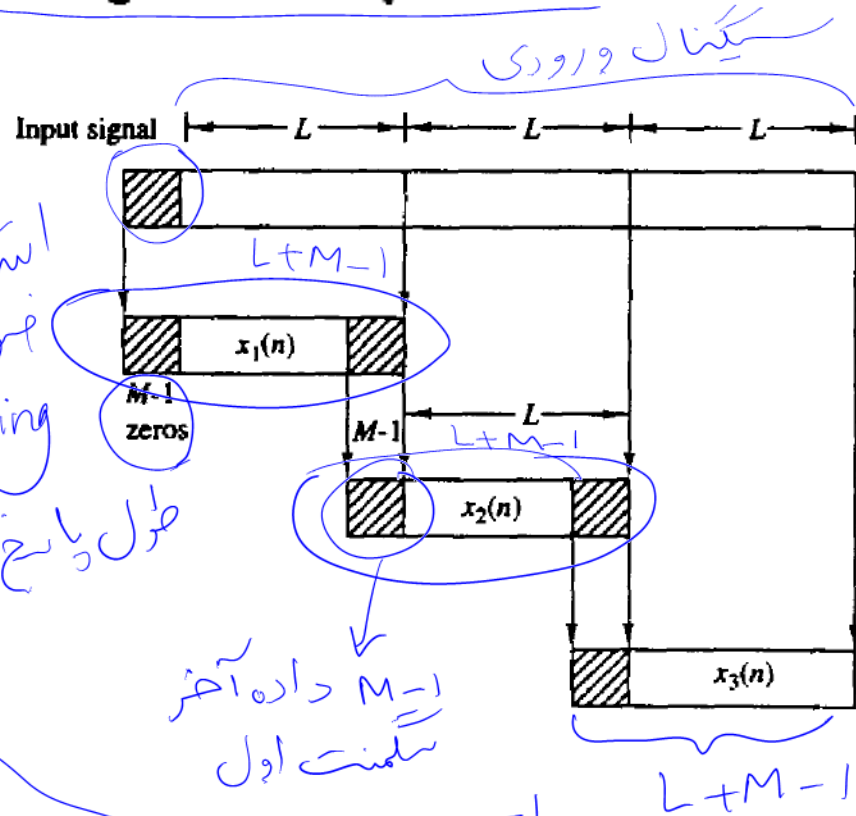
N نمونه در حوزه فرکانس

$$N \geq L + M - 1$$

کانولوشن چرخشی  
برابر خواهد بود  
کانولوشن خطی

# Filtering of Long Data Sequences

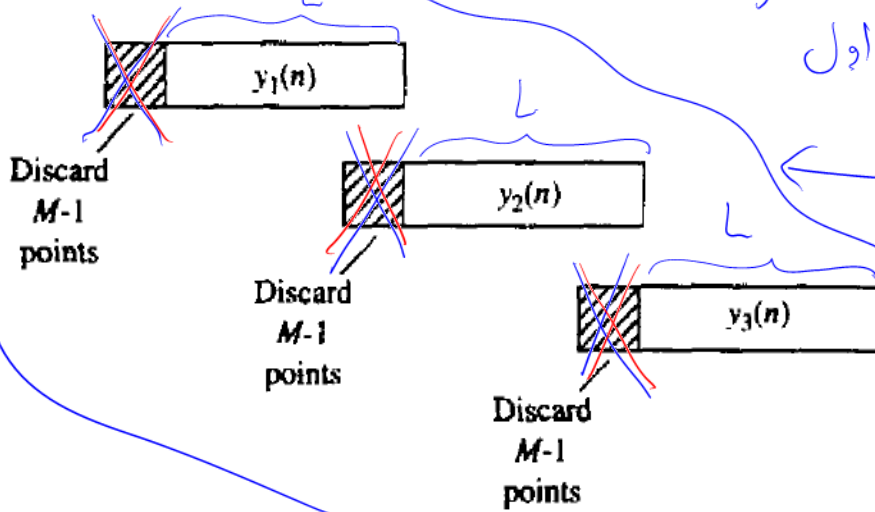
## Overlap-save method.



ابتدای سگمنت (بخش اول)  
 فیلتر اضافی و سگمنت  
 zero-padding  
 طول پاسخ ضرب (مقایسه)

$M-1$  داده آخر  
 سگمنت اول

Output signal

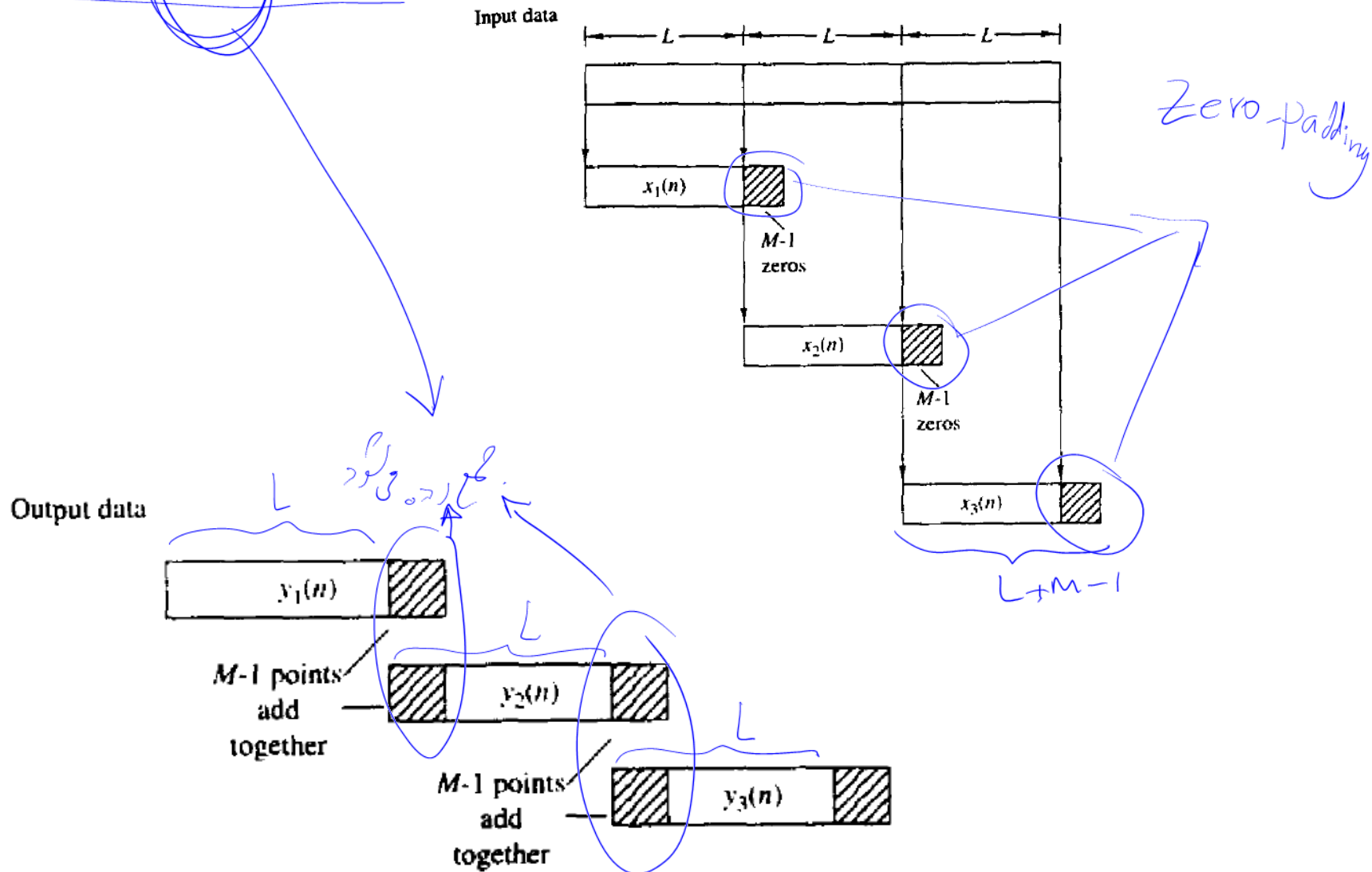


$DFT^{-1}$

$$Y[k] = X[k] \cdot H[k]$$

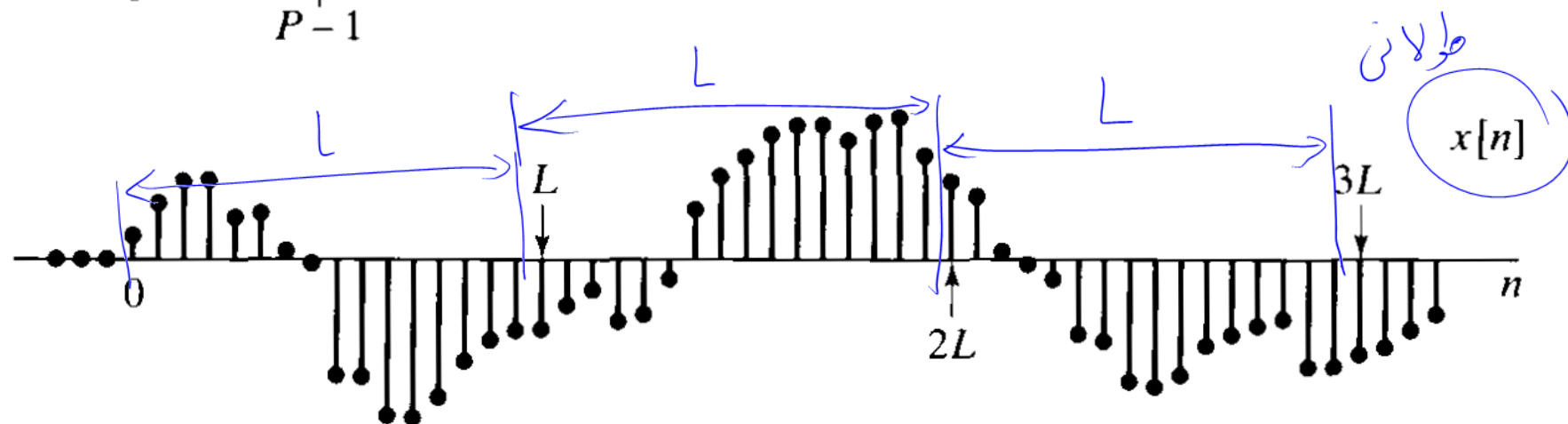
خروجی  $y[n]$

# Overlap-add method.

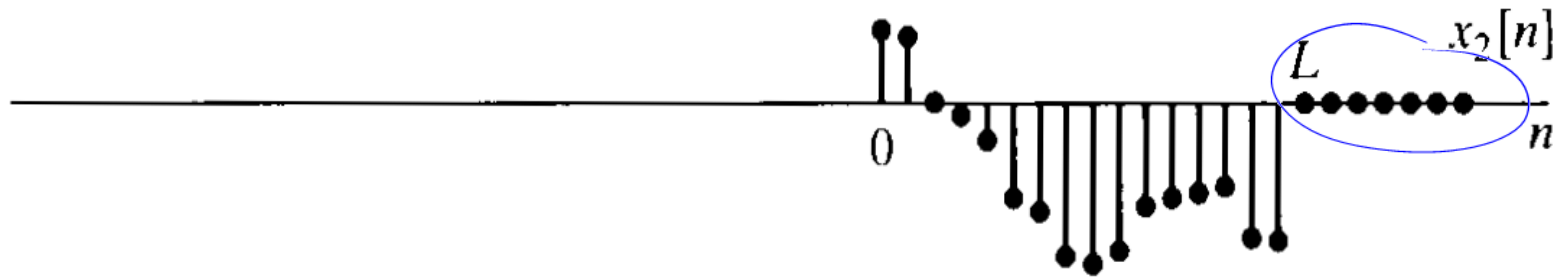
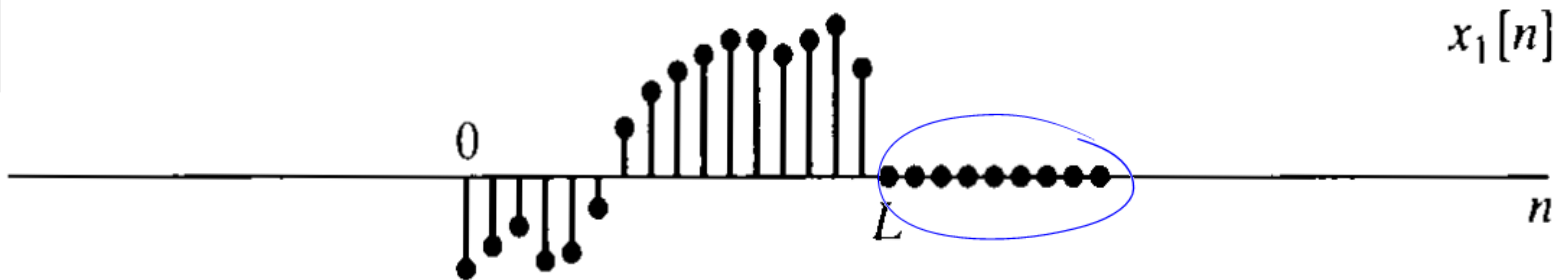
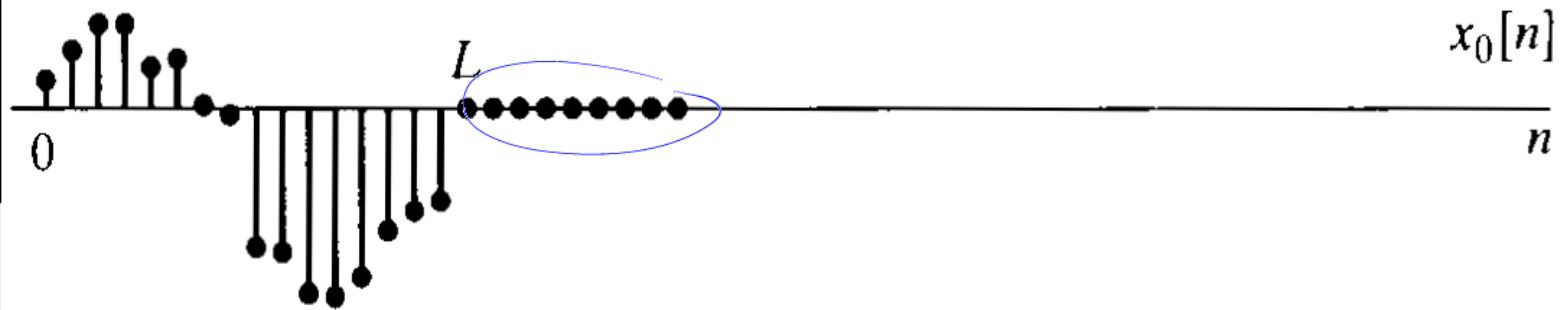


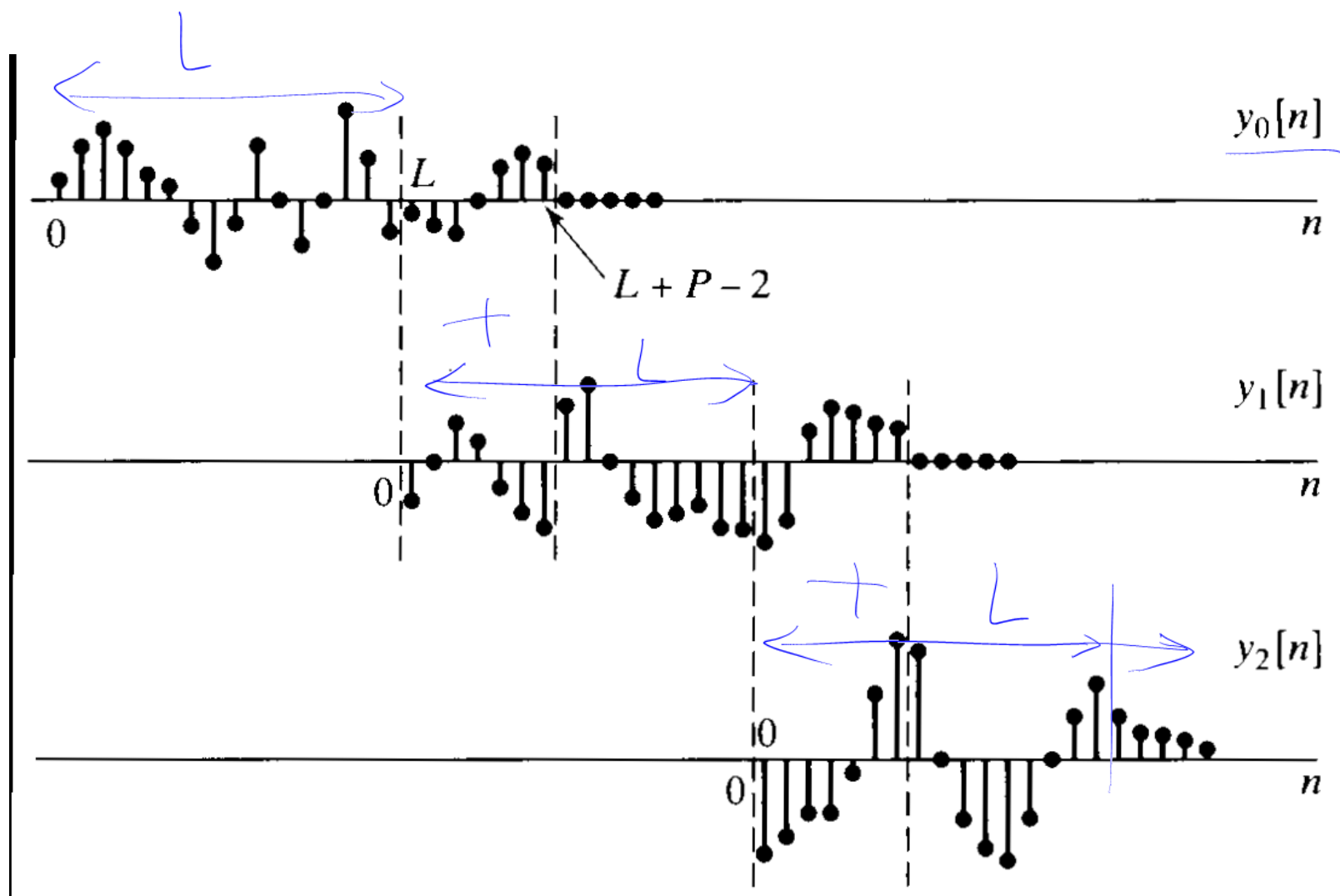
$$P=M$$

$$h[n]$$

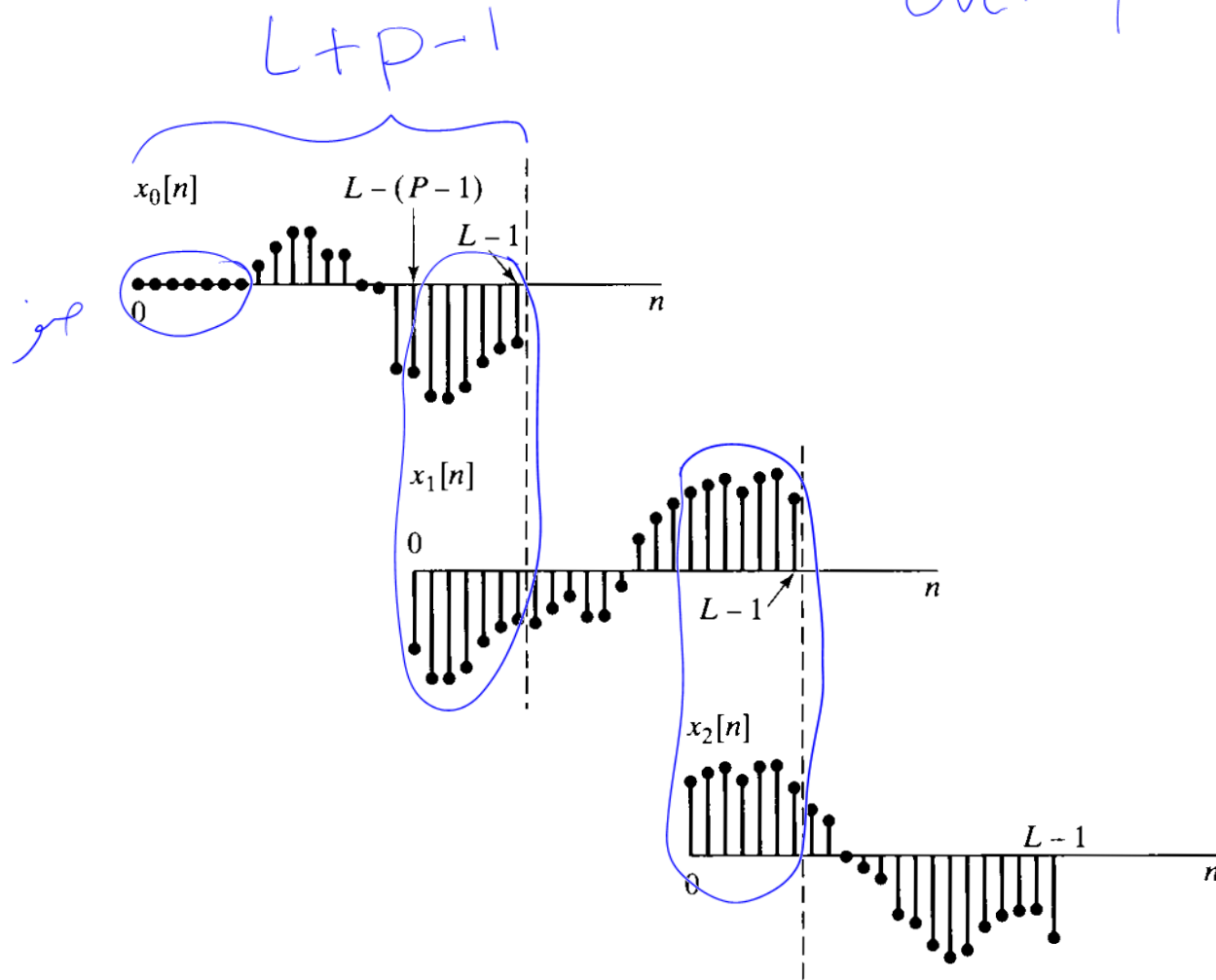


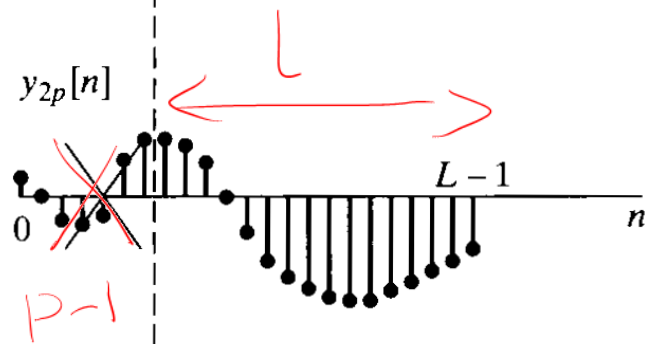
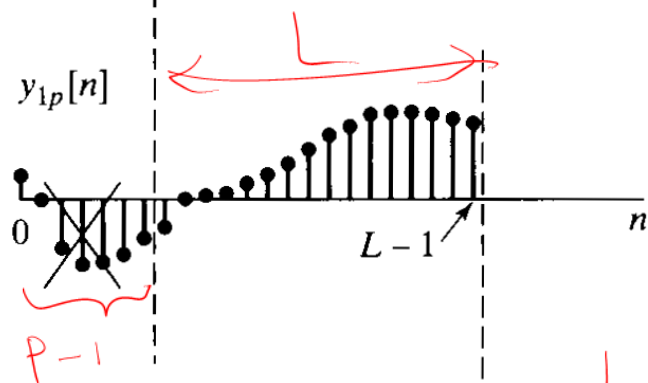
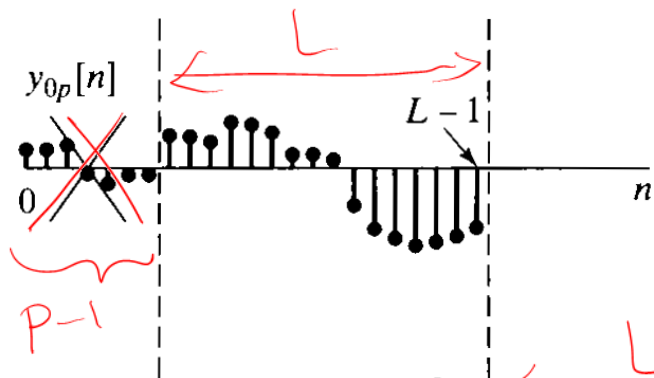
overlap-add





overlap-save







تابع یاب فوریه

(کرنل)

## Direct Computation of the DFT

$$X[k] = X_R[k] + jX_I[k]$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad k = 0, 1, \dots, N-1,$$

$$e^{j\frac{2\pi}{N}kn} = \cos\frac{2\pi}{N}kn + j\sin\frac{2\pi}{N}kn$$

$$(a+jb)(c+jd)$$

$$X_R(k) = \sum_{n=0}^{N-1} \left[ x_R(n) \cos \frac{2\pi kn}{N} + x_I(n) \sin \frac{2\pi kn}{N} \right]$$

$= (ac - bd) + j(ad + cb)$   
 حقیقی      موهومی

$$X_I(k) = - \sum_{n=0}^{N-1} \left[ x_R(n) \sin \frac{2\pi kn}{N} - x_I(n) \cos \frac{2\pi kn}{N} \right]$$

sin      cos

تایید می‌شود

1.  $2N^2$  evaluations of trigonometric functions.

2.  $4N^2$  real multiplications.

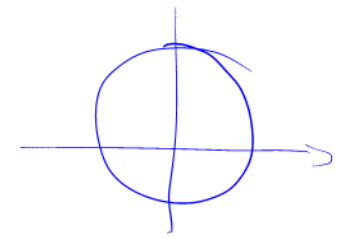
3.  $N(4N-2)$  real additions.

4. A number of indexing and addressing operations.

$N^2$

1965

Tuckey & Cooley



$$e^{-j\frac{2\pi}{N}kn}$$

# FFT method for computing DFT

Fast Fourier Transform

$$W_N^{k[N-n]} = W_N^{-kn} = (W_N^{kn})^*$$

$$W_N^{kn} = W_N^{k(n+N)} = W_N^{(k+N)n}$$

تقارن مزدوج مختلط  
(complex conjugate symmetry);

(periodicity in  $n$  and  $k$ ).

متناوب بودن در زمان و فرکانس

$$N = 2^V$$

طول سیگنال بصورت توان ۲ است

## Decimation in time (Radix-2)

فقط در زمان متناوب است

$N/2$  نقاطی DFT

$$X[k] = \sum_{r=0}^{(N/2)-1} x[2r] W_N^{rk}$$

$$+ W_N^k \sum_{r=0}^{(N/2)-1} x[2r+1] W_N^{rk}$$

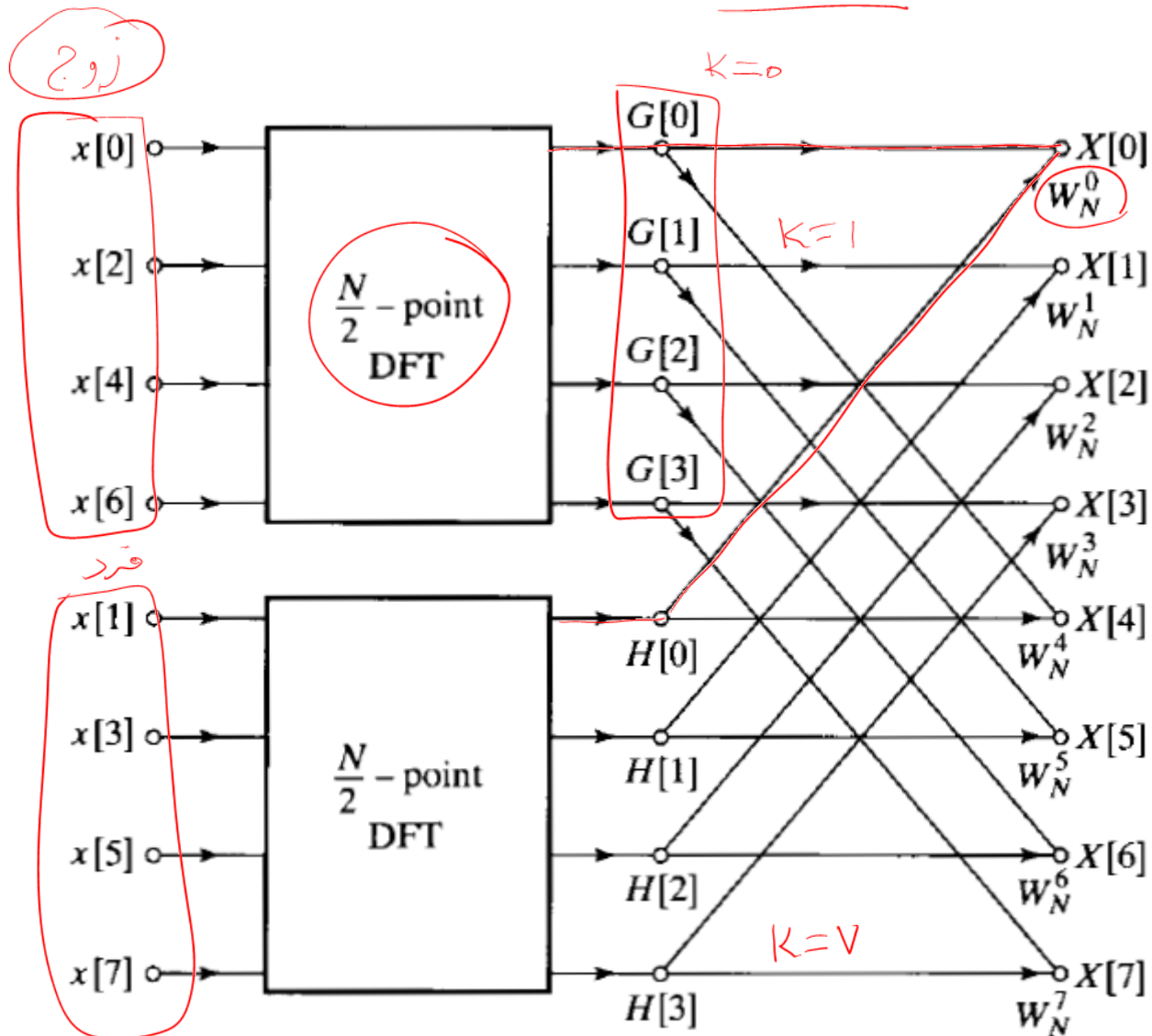
$$X[k] = G[k] + W_N^k H[k]$$

$$k = 0, 1, \dots, N-1.$$

$$n < \begin{matrix} 2r \\ 2r+1 \end{matrix}$$

# An example with 8-point

$$N = 8 = 2^3$$

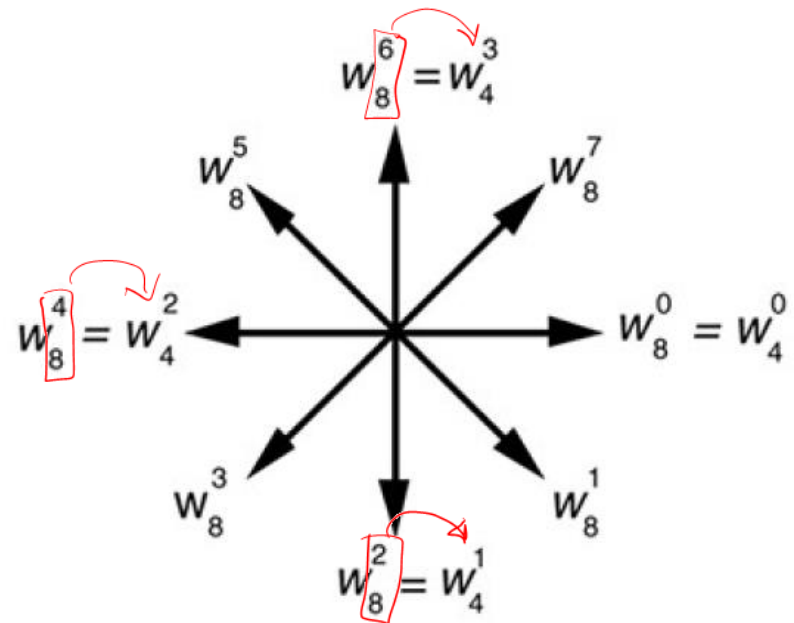
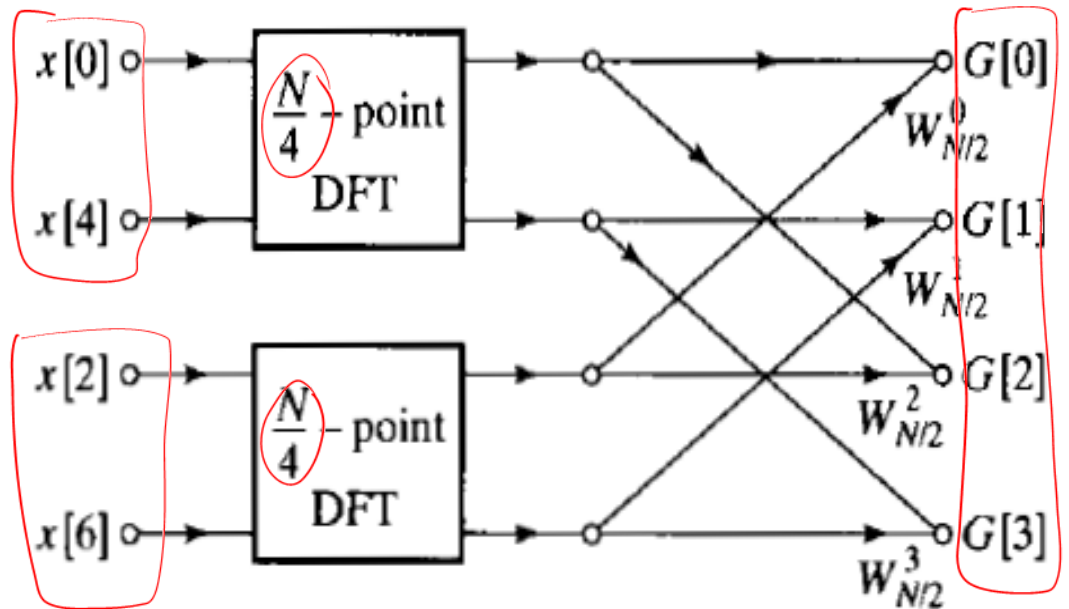


(زوج مرصده اول)

$$G[k] = \sum_{\ell=0}^{(N/4)-1} g[2\ell] W_{N/4}^{\ell k} + W_{N/2}^k \sum_{\ell=0}^{(N/4)-1} g[2\ell+1] W_{N/4}^{\ell k}$$

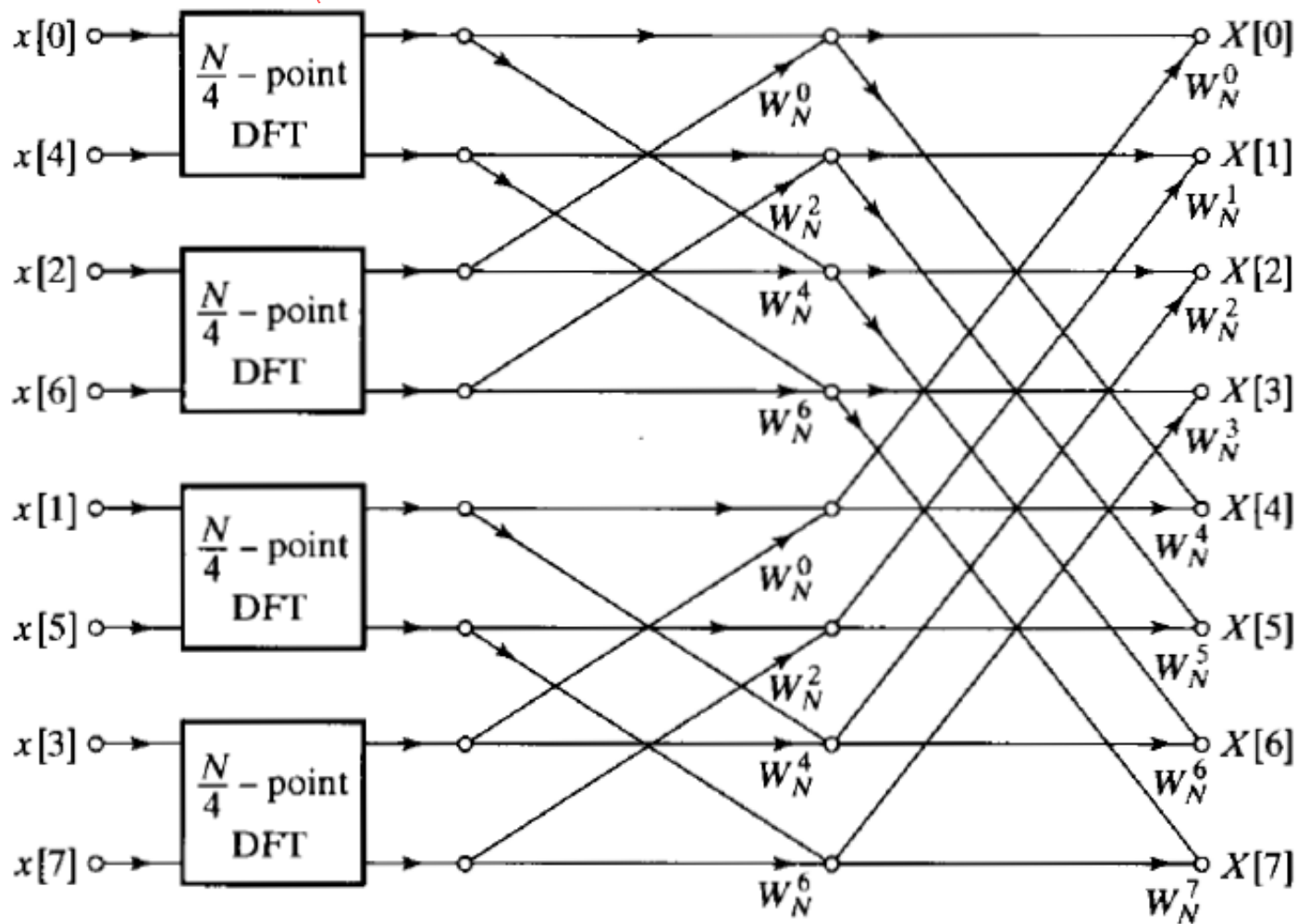
$$H[k] = \sum_{\ell=0}^{(N/4)-1} h[2\ell] W_{N/4}^{\ell k} + W_{N/2}^k \sum_{\ell=0}^{(N/4)-1} h[2\ell+1] W_{N/4}^{\ell k}$$

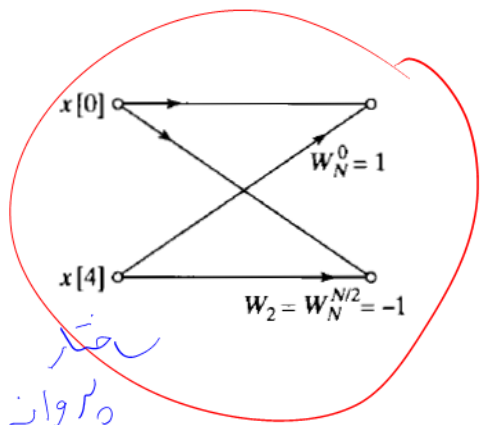
$N=8$



$$N - 1 \rightarrow N/4 = r$$

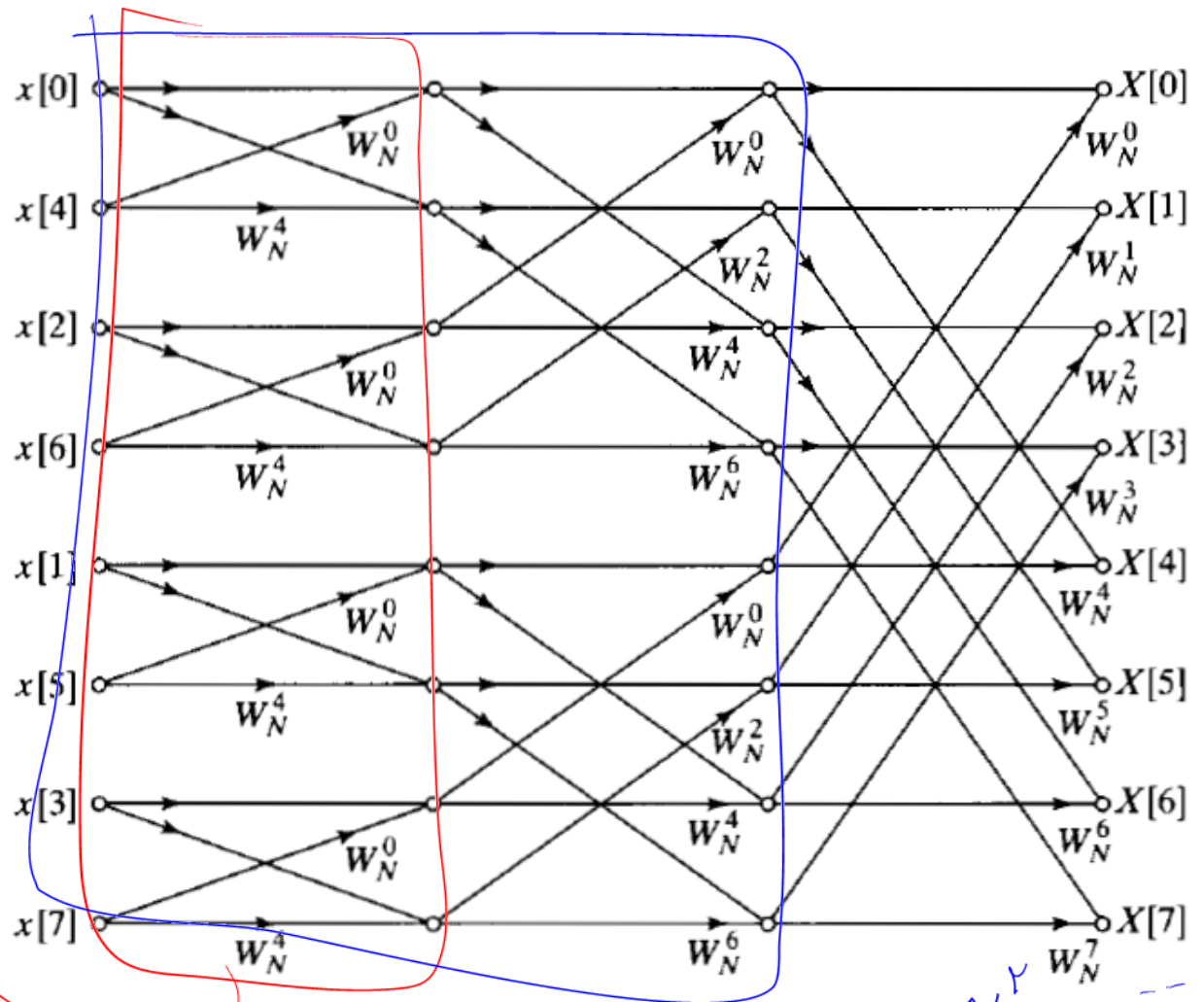
بقية





ساخت  
درونمای

DFT دو نقطه ای



نقطه ای

سقیم  $N^2$

$N = 2^v$   
 $v = \log_2 N$

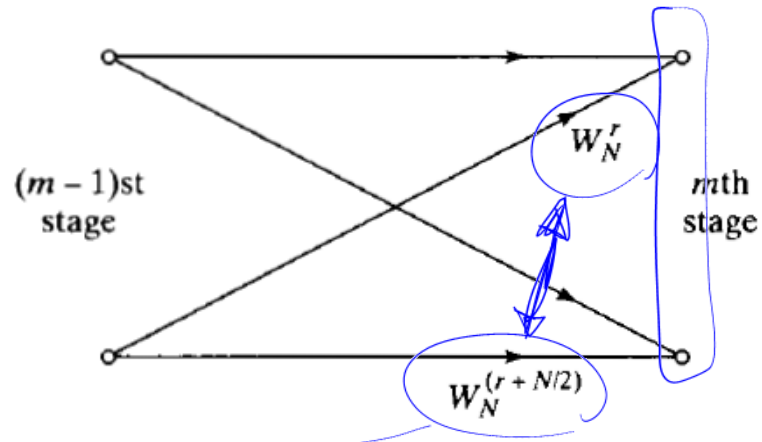
complex multiplications and additions is equal to  $Nv = N \log_2 N$ . FFT

$N = 2^{10}$   
 ۱۰۲۴

$N^2 = 2^{20} = 1,048,576$ , and  $N \log_2 N = 10,240$ . FFT

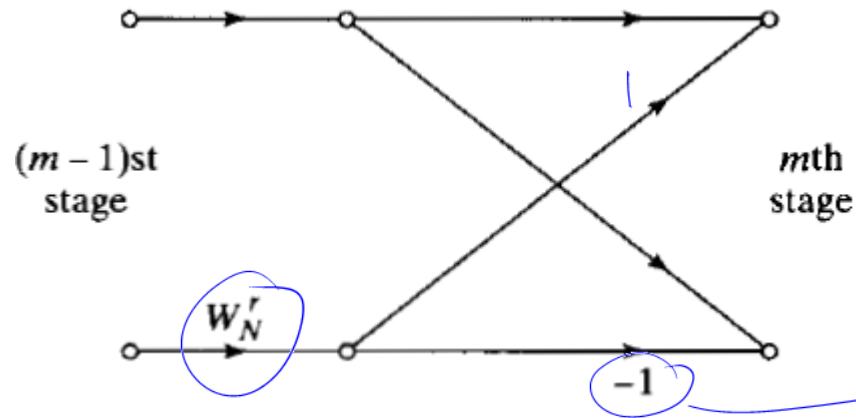
$$W_N^{N/2} = \left( e^{j\frac{2\pi}{N}} \right)^{N/2}$$

$$= e^{j\frac{2\pi}{N} \cdot \frac{N}{2}} = e^{j\pi} = -1$$



$$W_N^{r+N/2} = W_N^{N/2} W_N^r = -W_N^r$$

$$W_N^{N/2} = e^{-j(2\pi/N)N/2} = e^{-j\pi} = -1$$



تغییر بیت  
sign

# Bit-reversed input

FFT

$x[n]$

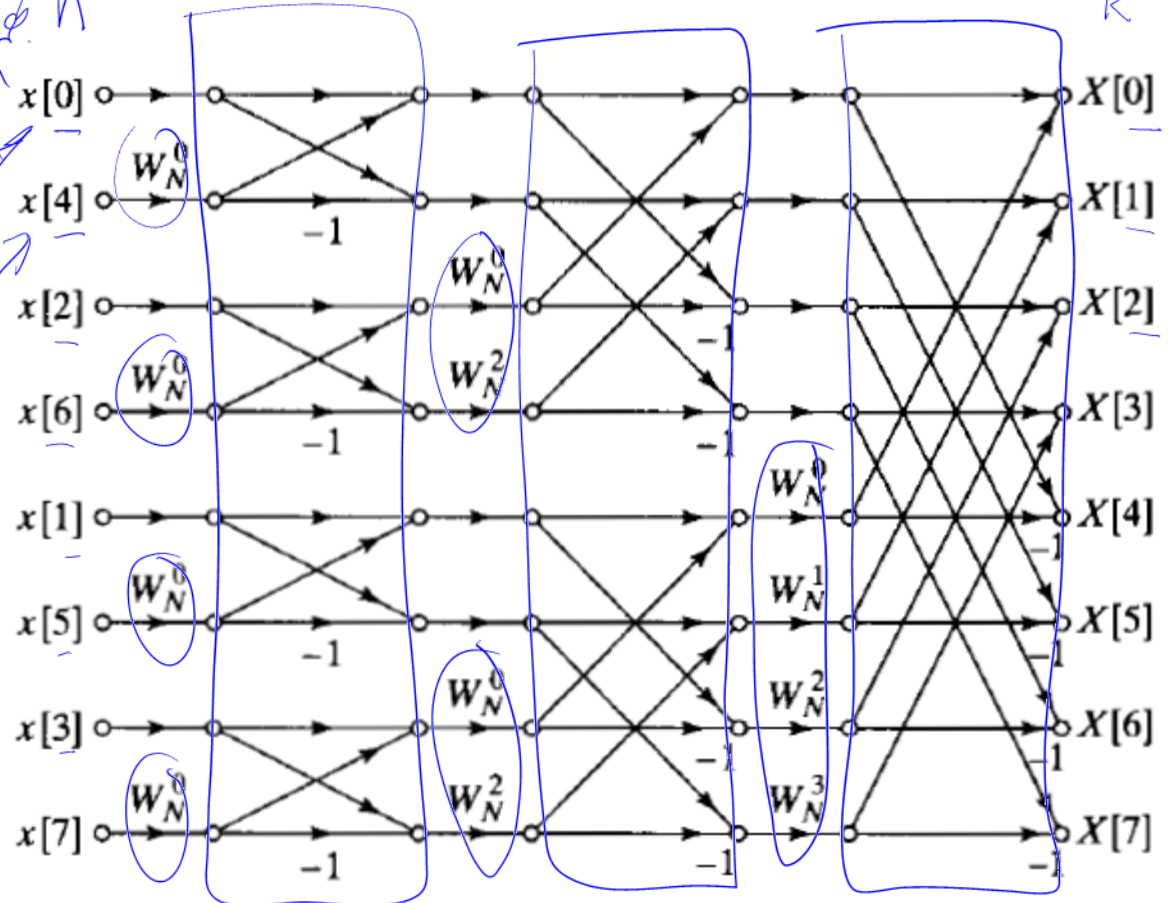
$X[k]$

Input order

Input sequence

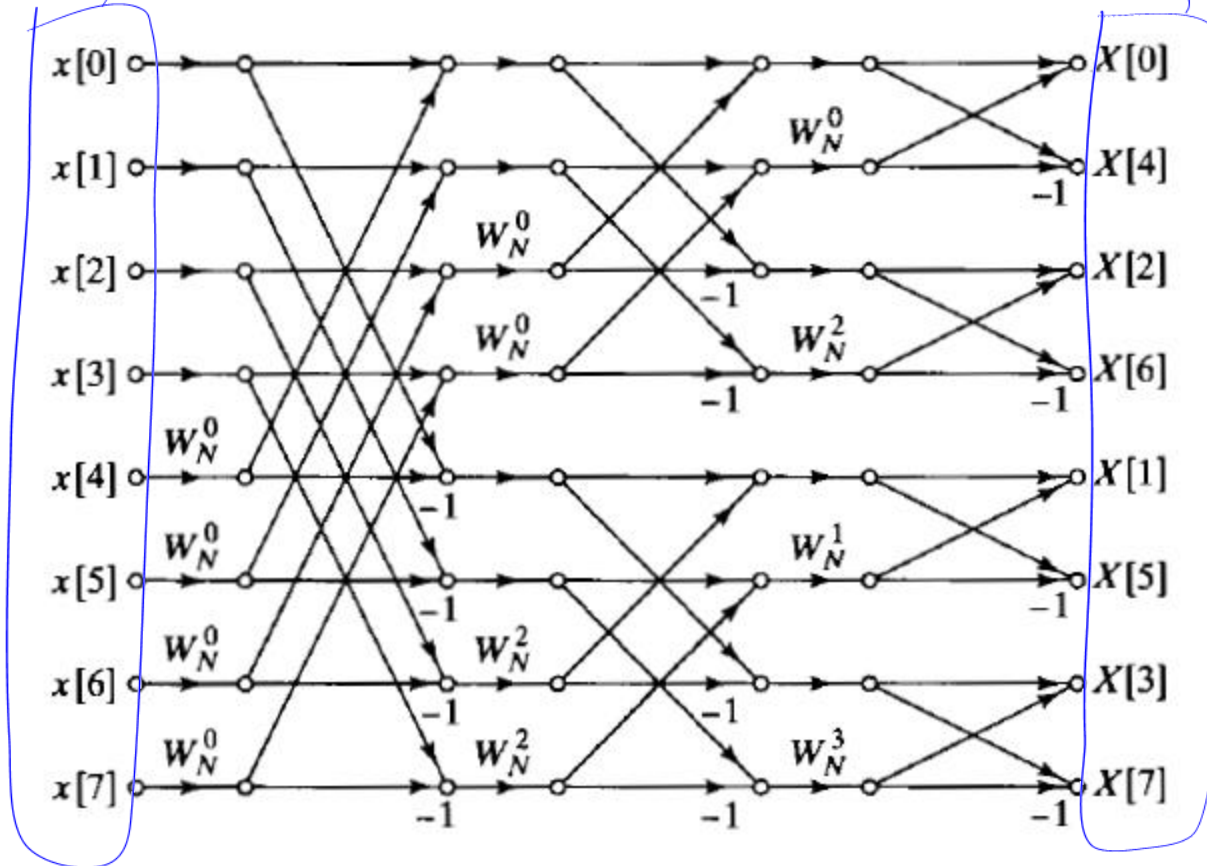
$X_0[n_0, n_1, n_2]$   $x[n_2, n_1, n_0]$

$X_0[000] = x[000],$   
 $X_0[001] = x[100],$   
 $X_0[010] = x[010],$   
 $X_0[011] = x[110],$   
 $X_0[100] = x[001],$   
 $X_0[101] = x[101],$   
 $X_0[110] = x[011],$   
 $X_0[111] = x[111].$

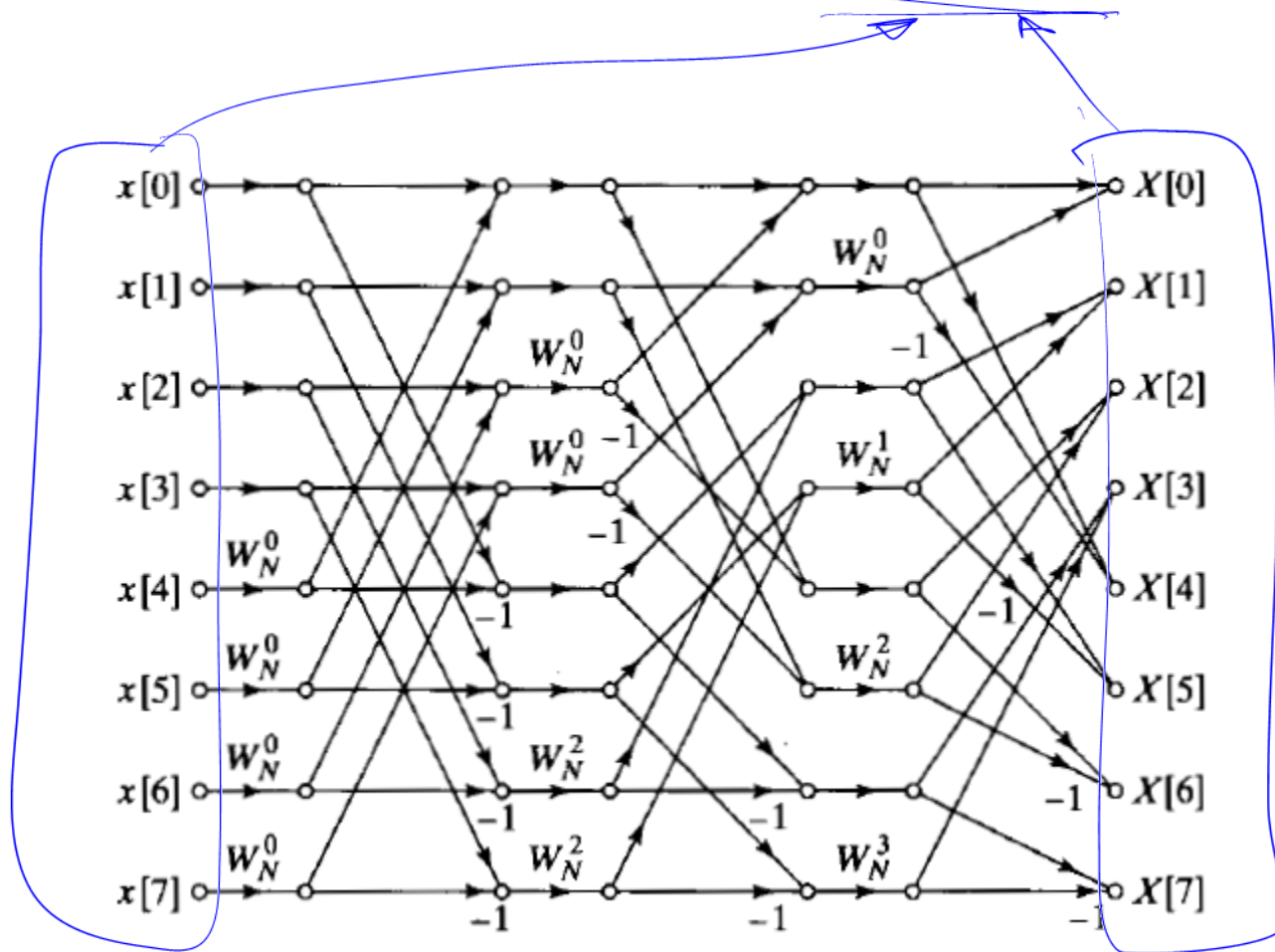




Input in normal order and output in bit-reversed order



Both input and output in normal order



Input in normal order and output in bit-reversed order  
and the same geometry for each stage

Modular

